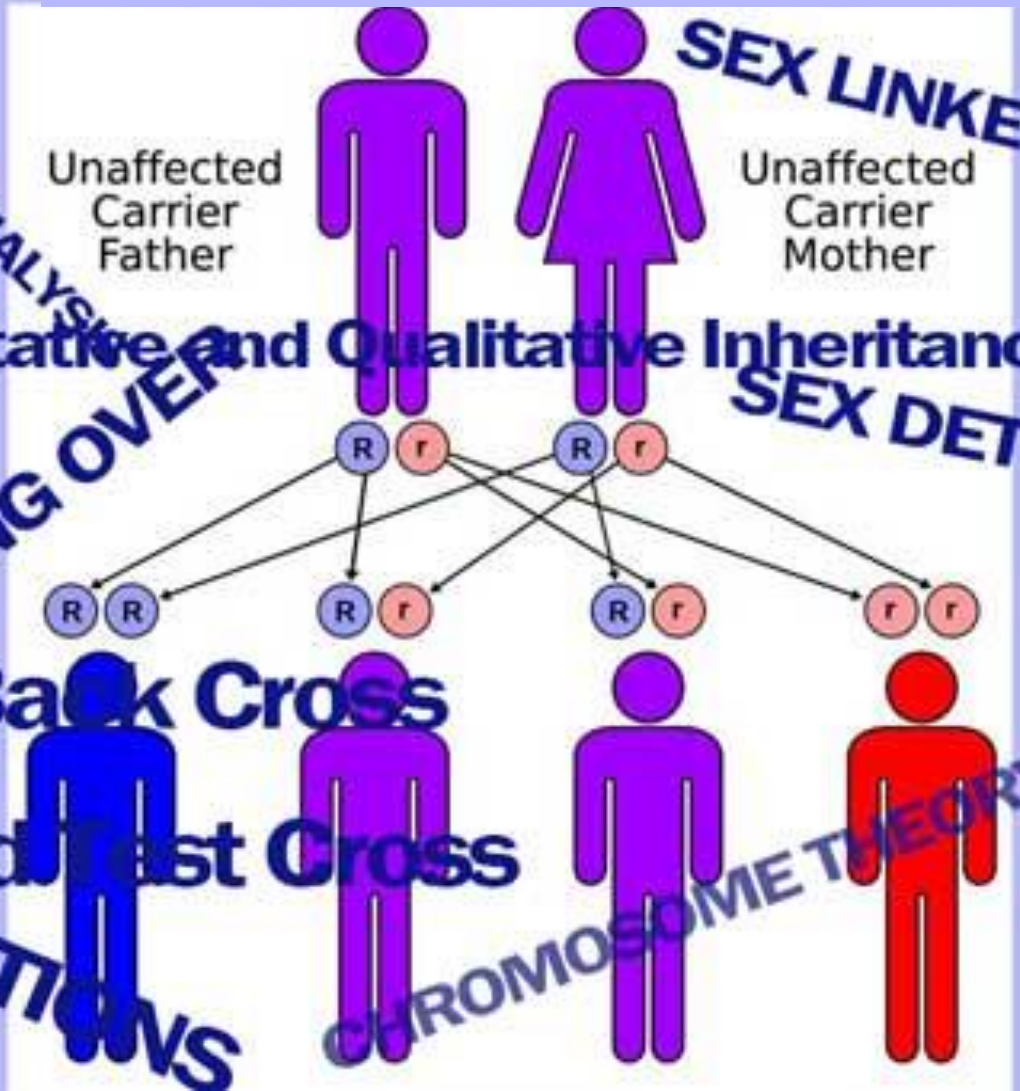
Principles of Inheritance and Variation



PEDIGREE ANALYSIS

SEX LINKED INHERITANCE

LINKAGE

Quantitative and Qualitative Inheritance

CROSSING OVER

SEX DETERMINATION

Back Cross

and Test Cross

CHROMOSOME THEORY OF INHERITANCE

VARIATIONS

MUTATION

What is inheritance?

- **Inheritance refers to the passage of hereditary traits one generation to the next.** Children/offspring resemble their parents because they inherit traits passed down from both parents.

- The branch of biology that deals with inheritance is called genetics.








Gregor Johann Mendel: Father of Genetics

- 1822- 1884
- Austrian monk
- Hybridization Experiment with pea plants.
- Published his results "Translation of the characters" in the natural history of society of brunn
- He is regarded as the 'father of genetics'



Seven pair of contrasting characters selected by mendel for his experiment.

	Height	Seed Shape	Seed Color	Seed Coat Color	Pod Shape	Pod Color	Flower Position
Dominant	 Tall	 Round	 Yellow	 Green	 Inflated (full)	 Green	 Axial
Recessive Trait	 Short	 Wrinkled	 Green	 White	 Constricted (flat)	 Yellow	 Terminal

GENES

- **"Genes are the units of inheritance which contain the information for the expression of a particular trait in an organism."**

Alleles

- Genes which code for a pair of contrasting traits are known as alleles. ie they are slightly different forms of the same gene .

Some terminologies in principles of inheritance

- **Homologous chromosomes:** The nuclei of all human cells except gametes contain 23 pairs of chromosomes-the diploid number (2n). One chromosome in each pair came from the mother and the other came from the father.
- **Gene locus:** the location of a gene on a chromosome, or within a DNA molecule, is called gene locus.

For example: “T” locus ,means **gene T** location on a chromosome. Gene T codes for a functional protein or enzyme.

- **Alleles**: A DNA sequence at the ‘T’ locus that has undergone mutation may no longer encode a functional enzyme. Following a mutation, the locus remains same, but the DNA sequence at mutant locus is different. So, different DNA sequences at the same locus are called different alleles of the same gene. Any change in the DNA sequence of gene creates new allele.

- **Example to understand the concept of alleles:**

Trait : Color of flower

Purple flower (P)

Purple pigmentation due to anthocyanin

Purple color is formed due to a specific functional enzyme

DNA sequence at "P locus" for the functional enzyme for production of anthocyanin

White flower (p)

No pigmentation, absence of anthocyanin

No functional enzyme

DNA sequence at "P locus" mutated for the functional enzyme for production of anthocyanin. **mutated version is now called here as "p" allele.**

- **Genotype**: genetic composition of an organism for the particular trait.
- **Phenotype**: The external appearance of an organism due to its genetic composition/influence of genes and environmental factors.

Phenotype

- The external appearance of an organism is called phenotype
- Eg . tall , dwarf , red flower , violet flower , round seed , wrinkled seed, axial flower ,
- terminal flower etc.

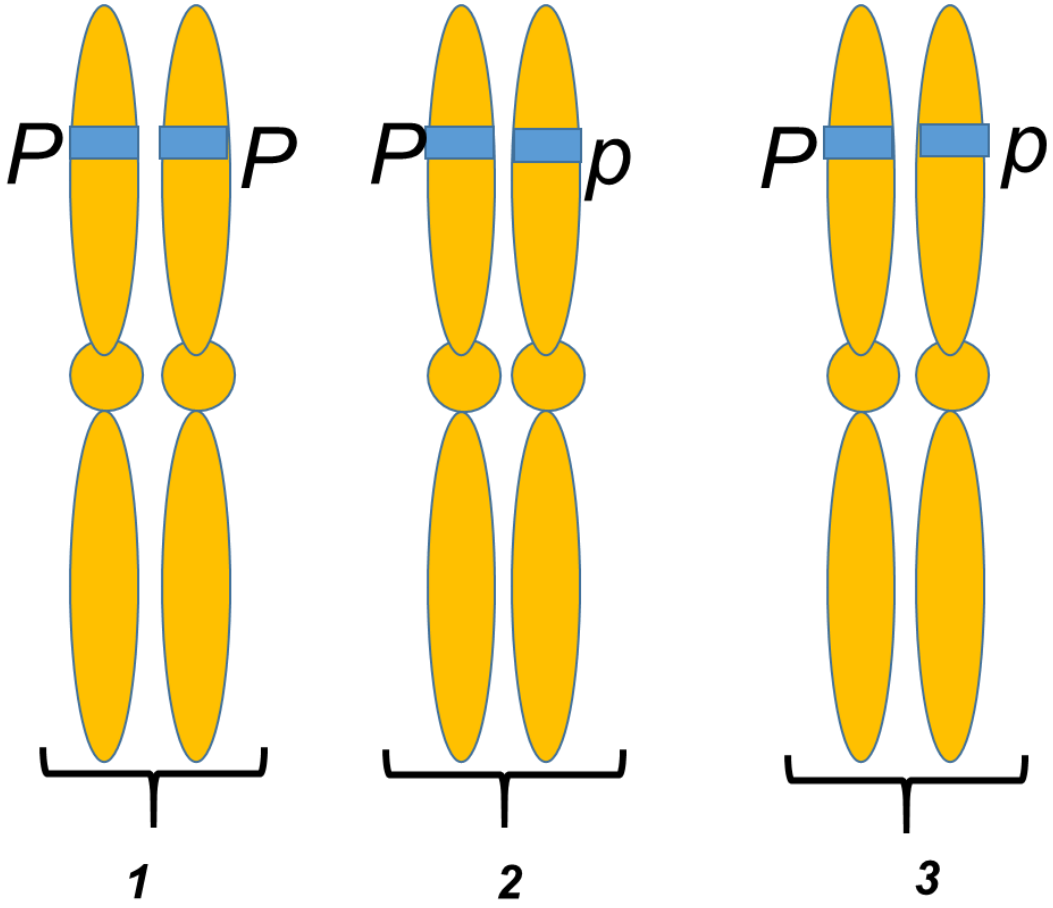
Genotype

- The genetic make up of an organism is the genotype.
- Eg ; TT and Tt express the tall genotype.

- **Homozygous**: the individuals having identical alleles in an allelic pair for a character or same alleles of a gene at the gene locus is called homozygous condition.
- **Example : "*PP*" for purple color and "*pp*" for white color.**
- **Heterozygous:** the individuals having different alleles in an allelic pair for a character or different alleles of a gene at the gene locus is called heterozygous condition.
- **Example : "*Pp*" for purple color.**
- **Dominant allele: If** out of two different allele of a gene at the locus, one allele is expressed, forms a functional product, and shows its phenotypic effect in every generation, then that allele is known **to be dominant over the other allele.**
- **For example, genetic make up of plant with *"Pp"* giving purple flower means *"P" allele* is dominant over *"p" allele.***
- ***Recessive allele:*** If out of two different allele of a gene at the locus, one allele is not expressed, or do not form any functional product, then that allele is known **to be recessive over the other allele**

Gene locus "P"

Three pairs of Homologous chromosomes



Genotype/ Genetic composition

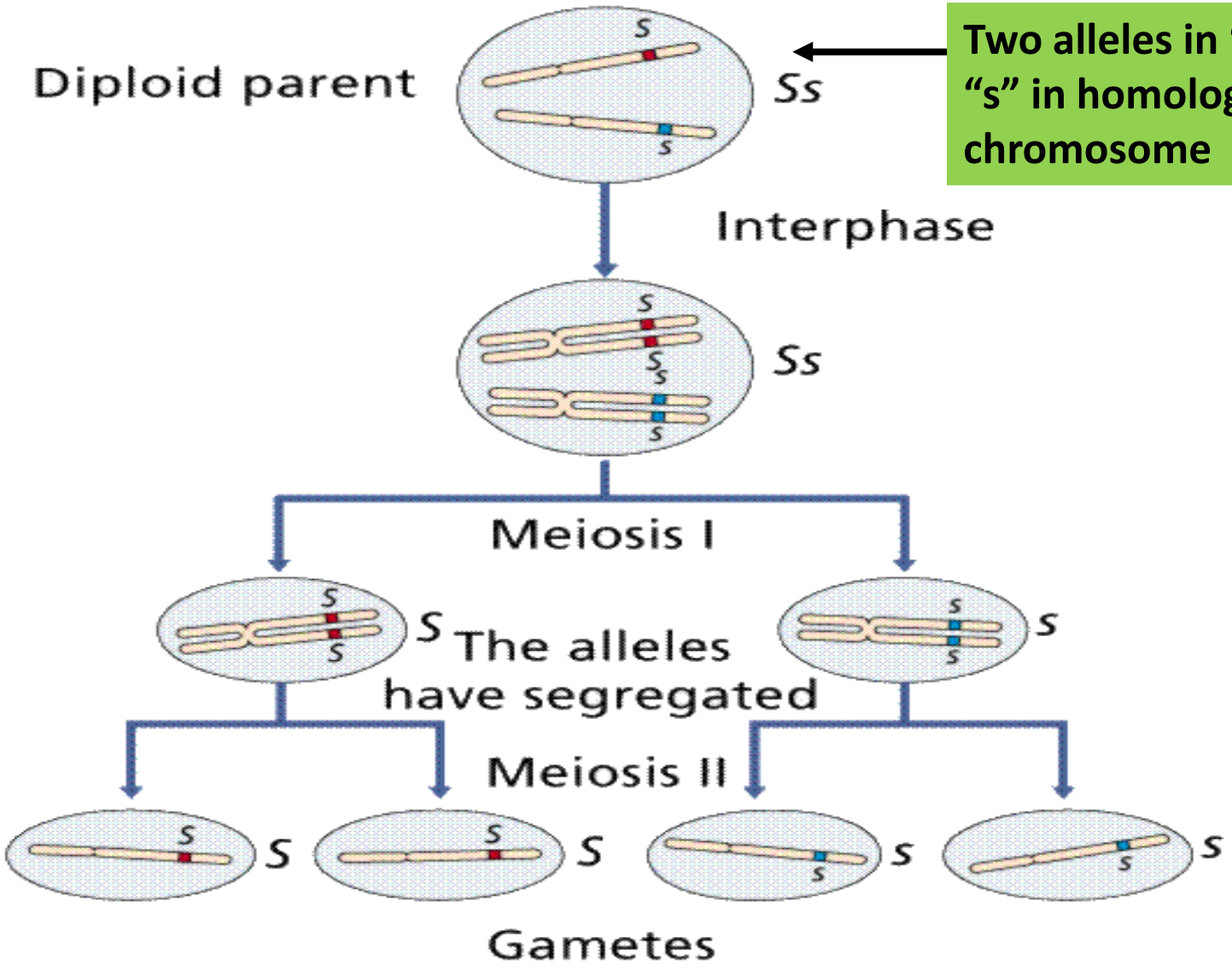
PP	***Pp***	***pp***
Homozygous	***Heterozygous***	***Homozygous***

Phenotype

Purple flower	***Purple flower***	***White flower***

Allelic pairs. *There are three allelic pairs for gene "P" at the gene locus*

How are alleles separated during gamete formation?



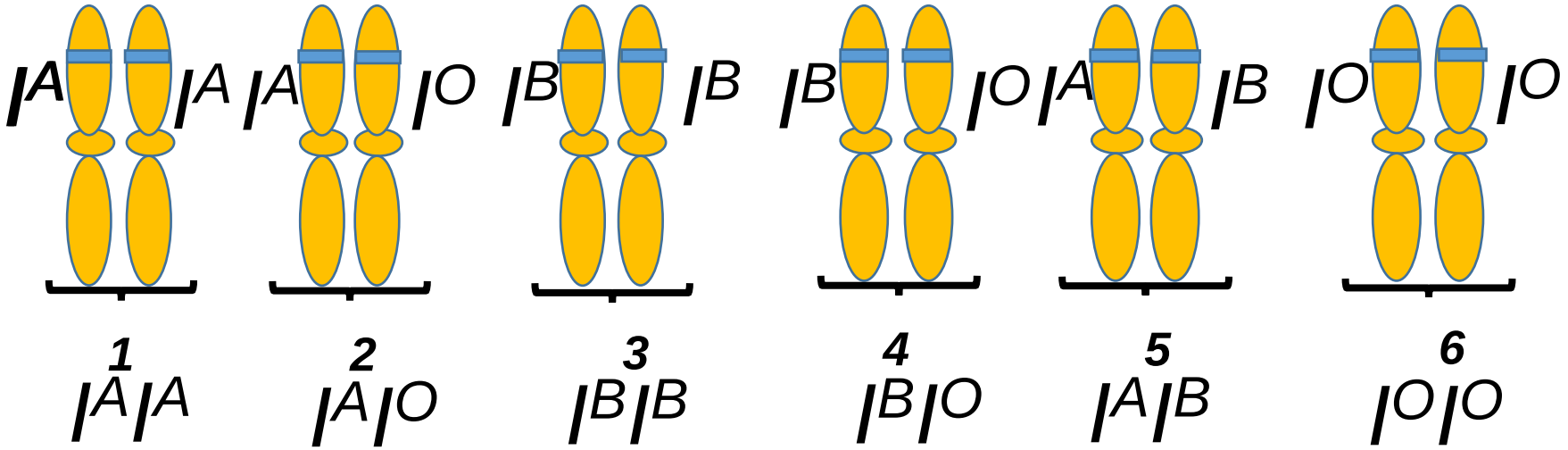
Codominance/multiple-allele inheritance

- **Codominance**: Both the alleles at the gene locus shows its independent affect. “ABO” –locus that determines the blood group of an individual.
- ABO locus has three major alleles- ABO*A; ABO*B; ABO*O. Also represented as I^A , I^B , I^O .

❖ At “ABO” gene locus of an individual, two alleles are present and therefore, six genotypes are possible. **(A and B alleles are dominant while O is recessive)**

Blood type	Genotype	
A	I^A, I^O I^A, I^A	AO AA
B	I^B, I^O I^B, I^B	BO BB
AB	I^A, I^B	AB
O	$I^O\ I^O$	OO

Chromosomal location of ABO blood group alleles



BLOOD TYPE CHART

Blood Type (Phenotype)	Surface Molecules	Alleles (Genotype)
Type A	A	AA or Ao
Type B	B	BB or Bo
Type AB	A and B	AB
Type O	None	oo

BLOOD TYPES BABY

Mother's blood type \ Father's blood type	A	B	AB	O
A	A, O	A, B, AB, O	A, B, AB	A, O
B	A, B, AB, O	B, O	A, B, AB	B, O
AB	A, B, AB	A, B, AB	A, B, AB	A, B
O	A, O	B, O	A, B	O

Why blood group alleles are codominant and not dominant?

- Person having blood group A has **“A-antigen”** on its RBC’s surface,
- Blood group B has **“B-antigen” on its surface**,
- Blood group **AB has both “A and B antigens” expressed on RBC** surface,
- O has **“no antigens”** on RBC surface.
- The codominant effect of alleles **is due to the equal expression of A and B antigens on the RBC surface.**
- No allele is dominant over other or masks the expression of other. Therefore, blood group alleles show the codominance phenomenon

X-linked inheritance

- The human X chromosome contains many genes that are required in both sexes, whereas Y chromosome contains only a few genes, primarily genes related to maleness.
- About 200 traits have been estimated to be X-linked compared to a few, which are Y-linked.
- **The traits controlled by genes located on X chromosome are often called sex-linked genes, or X-linked traits.**
- Females have two doses of X-linked genes (because two X-chromosome) and males have only one dose.

Female genotype : XX; males genotype : XY

- One dose of **X-linked dominant allele** produces its effects in males as well as females

For example: X^HX X^HY; where "H" is the dominant allele and will show its effect on both males and females.

- **If the allele is X-linked recessive**, then female will show the phenotypic effect only when both X-chromosome has the allele, while males will show the phenotypic effect due to presence of only one X-chromosome
- **for example: X^hX X^hY; where "h" is the recessive allele** and will show its phenotypic effect only in males while females will be the carrier of recessive allele. **Females will show its effect only when both the X-chromosome carries the recessive allele.**

Inheritance pattern of X and Y

	Female	Male
Parental generation	44+XX	44+XY

Gametes: ovam in female and sperm in males

22+X 22+X 22+X 22+Y

Fertilization of Ovam and Sperm

44+XX 44+XY 44+XX 44+XY

Punnet square

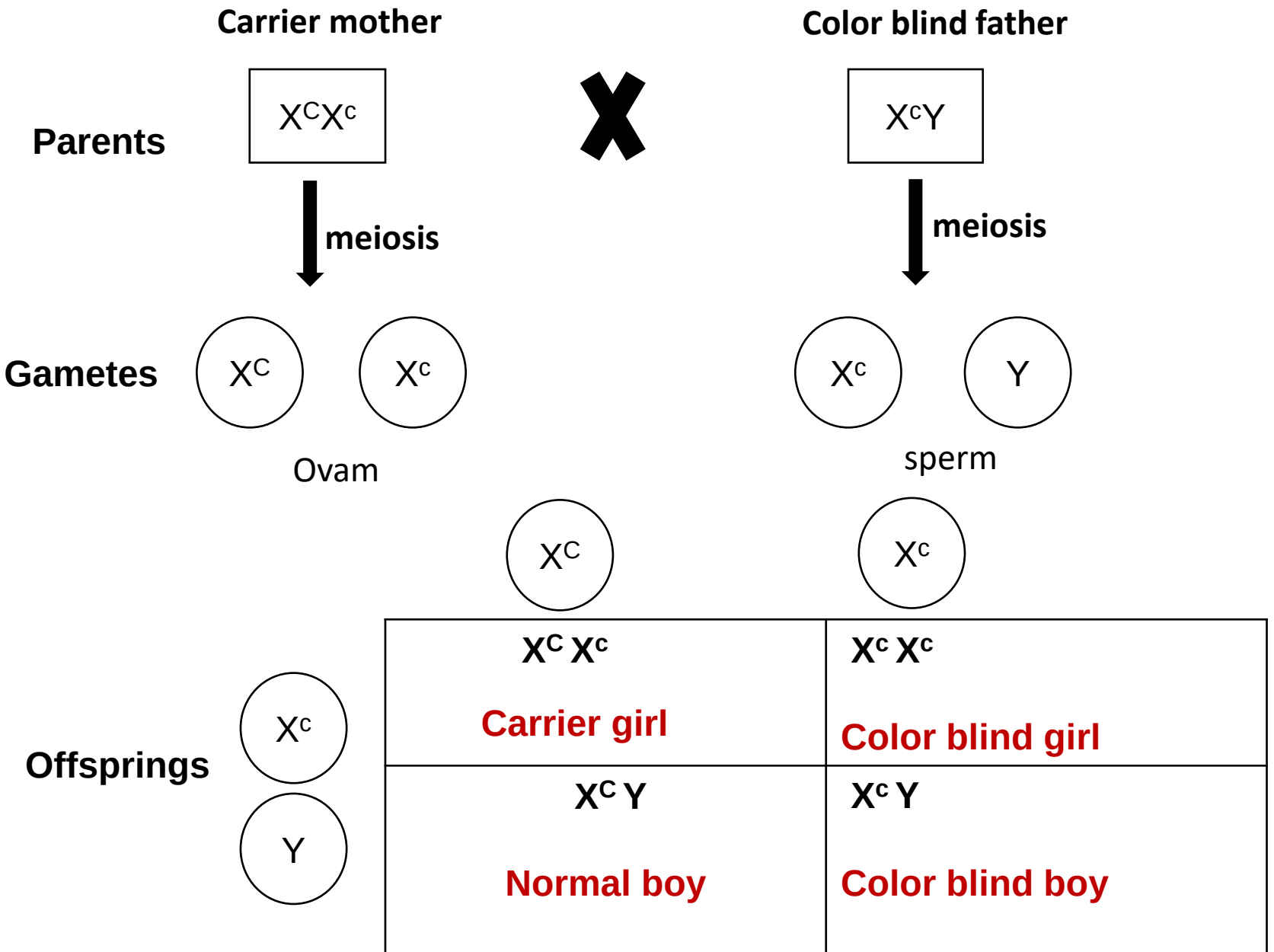
Female gamete → / Male gamete ↓	X	X
X	XX	XX
Y	XY	XY

Red-green color blindness and haemophilia is an example of X-linked recessive inheritance

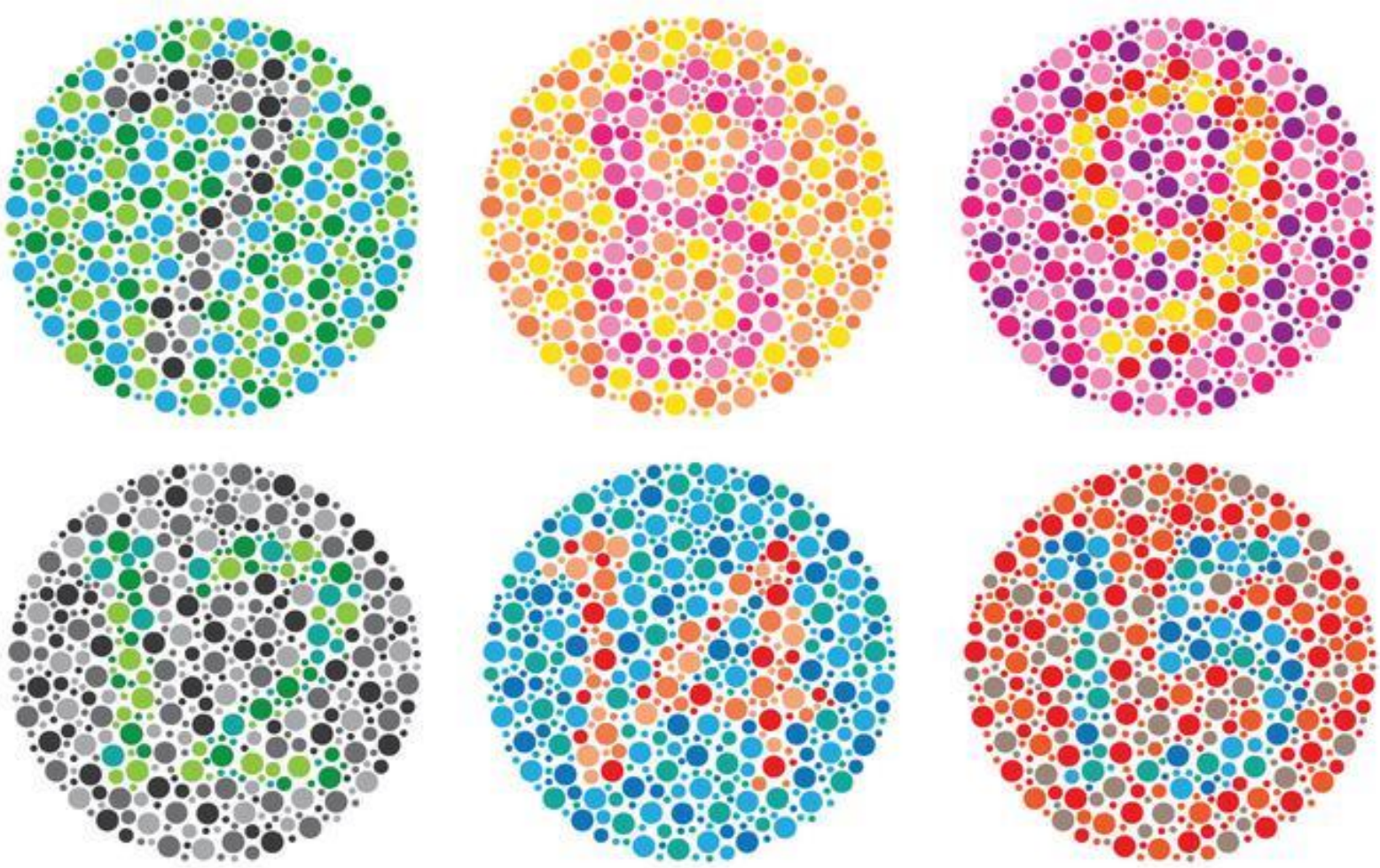
- Red-green color blindness condition is characterized by a deficiency in either red-or-green-sensitive cones.
- Therefore, red and green are seen as same color (either red or green, depending on which cone is present).
- The gene for red-green color blindness is a recessive one designated as “c”. Normal color vision , designated as “C” dominates.
- The C/c genes are located on X-chromosomes and thus the ability to see colors depends entirely on the X-chromosome. The possible combinations are as follows:

Genotype	Phenotype
X^CX^C	Normal female
X^CX^c	Normal female but carrier of recessive gene
X^cX^c	Red-green color-blind female
X^CY	Normal male
X^cY	Red-green color-blind male

Inheritance pattern of red-green color blindness



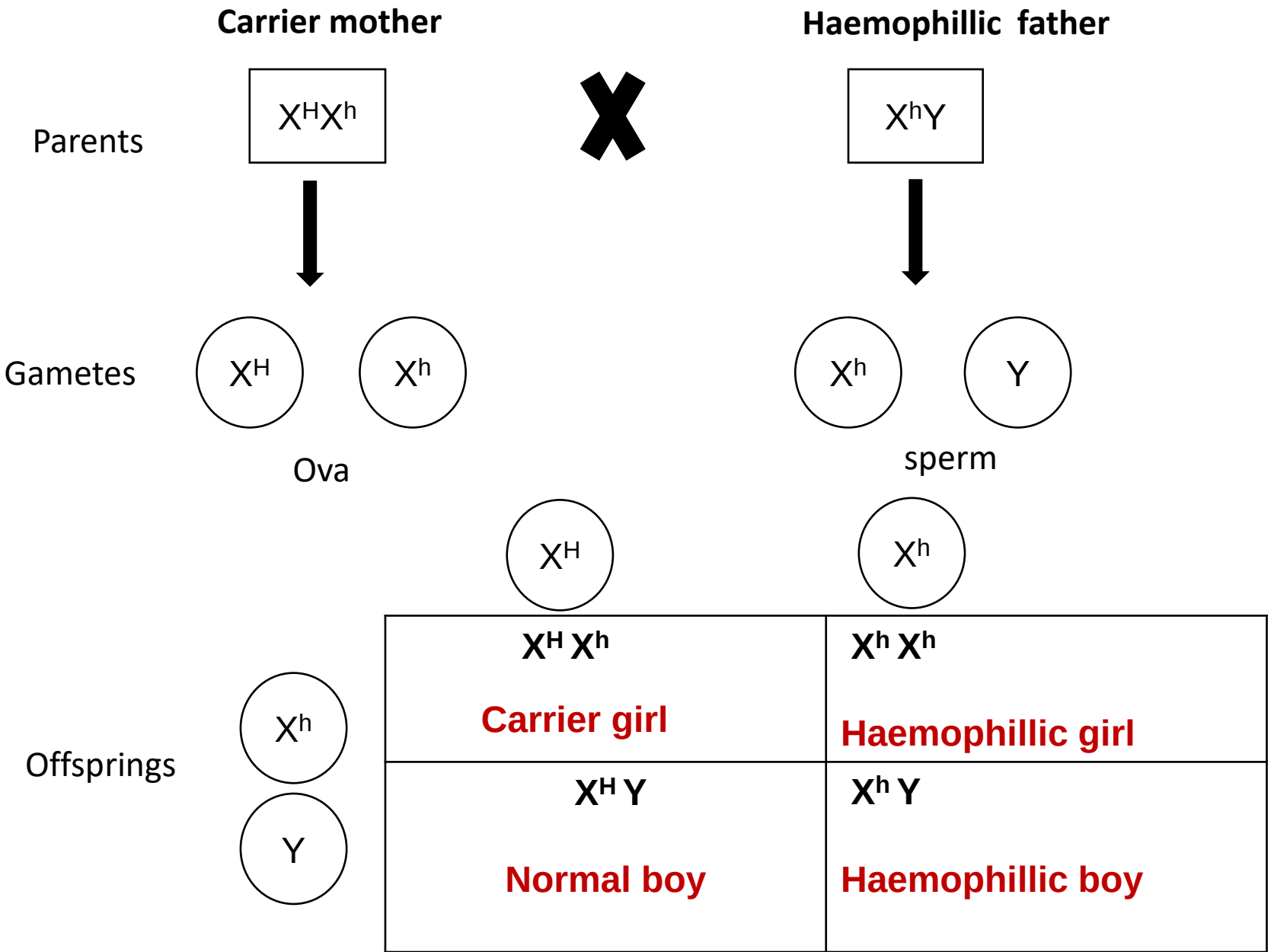
Color blindness test chart



- Haemophilia is also a sex-linked trait, in which the blood fails to clot or clots very slowly after an injury.

Genotype	Phenotype
X^HX^H	Normal female
X^HX^h	Normal female but carrier of recessive gene
X^hX^h	Haemophillic female
X^HY	Normal male
X^hY	Haemophillic male

Inheritance pattern of haemophilia



Sample questions

Q.1) A woman with two genes for haemophilia marries a normal man.

What will be the phenotypes of progeny? Choose from Options:

A) All sons and daughters are haemophillic

B) All normal sons

C) 100% chances of sons to be hemophilic and all daughters will be the carrier.

Q.2) A mother with blood type A has a child with blood type A. Give all possible blood types for the father of this child from the options below.

a) O

b) B, AB

c) A, AB

d) A, B, O

e) A, B, AB,O

Q.3) A mother of blood type A gives birth to a child with blood type O. Which of the following could NOT be the blood type of the father?

a) A

b) B

c) O

d) AB

e) Any of the above is a possible blood type of the father

Q.4) If a woman is a carrier for color-blind gene, marries a color-blind man, and have daughter, what is the percent chance that she will have color blindness?

???